A 4^r-Scaled Family of the Ternary $(4k - 1(2))/3$ Collatz-Type Map

Exact Algebraic Conjugacy, Cycle Preservation, and Transfer of the 10^9 Finite Verification

Farhad Banazadeh
fn.bana@hotmail.com
ORCID: 0009-0004-7023-0298

9 September 2026

Abstract

We develop the full 4^r-scaled family of the ternary $(4k - 1(2))/3$ Collatz-type map introduced and computationally studied in the author's preceding work. The base state space is

$$\mathcal{S} = \{y \in \mathbb{Z}_{>0} : 3 \nmid y\},$$

with residue selector $\rho(y) = y \bmod 3 \in \{1, 2\}$, affine numerator $M_0(y) = 4y - \rho(y)$, and complete 3-adic reduction

$$T_0(y) = \frac{4y - \rho(y)}{3^{\nu_3(4y-\rho(y))}}.$$

For every integer $r \geq 0$ we define the scaled domain $\mathcal{S}_r = 4^r\mathcal{S}$ and the scaled affine map

$$M_r(x) = 4x - 4^r\rho(x), \qquad x \in \mathcal{S}_r,$$

so its two branches are $4x - 4^r$ and $4x - 2 \cdot 4^r$. After complete removal of all factors of 3, the scaled compressed map is $T_r(x) = M_r(x)/3^{\nu_3(M_r(x))}$.

The central result is exact algebraic conjugacy. Since $4^r \equiv 1 \pmod 3$ and $3 \nmid 4^r$, multiplication by 4^r preserves both the branch residue and the 3-adic valuation. Consequently

$$T_r(4^r y) = 4^r T_0(y), \qquad T_r^j(4^r y) = 4^r T_0^j(y) \quad (j \geq 0).$$

Thus every orbit relation, periodic orbit, cycle length, valuation sequence, transient length, and basin membership is transported bijectively between the base system and its scaled copy; absolute peak values are multiplied by 4^r, while normalized peak ratios are unchanged. In particular, the base fixed points 1 and 2 become 4^r and $2 \cdot 4^r$, and the base four-cycle $22 \to 29 \to 38 \to 50 \to 22$ becomes

$$22 \cdot 4^r \to 29 \cdot 4^r \to 38 \cdot 4^r \to 50 \cdot 4^r \to 22 \cdot 4^r.$$

No additional cycle can occur inside $\mathcal{S}_r$: dividing any scaled cycle by 4^r gives a base cycle of the same length.

The preceding base paper exhaustively verified all $666,666,667$ admissible starts through 10^9. We do not repeat that billion-scale computation here. Instead, the conjugacy theorem transfers every one of those verified trajectories exactly to the corresponding set $\{4^r y : y \leq 10^9,\ 3 \nmid y\} \subset \mathcal{S}_r$, extending through $4^r 10^9$ in absolute scale. The global three-attractor convergence statement remains a conjecture: the present theorem proves exact equivalence of the scaled and base dynamical problems, not global convergence of the unresolved base problem.

Keywords: Collatz-type map; ternary dynamics; 4^r scaling; algebraic conjugacy; 3-adic valuation; periodic orbits; cycle preservation; computational number theory.

1 Introduction

The ternary $(4k-1(2))/3$ map studied in the author's preceding paper [2] is an accelerated integer dynamical system on positive integers not divisible by 3. At each state, multiplication by 4 is followed by subtraction of the nonzero residue modulo 3, and then by complete removal of all factors of 3. The base paper gave the exact residue branches, inverse branches, a Diophantine cycle equation, a 3-adic drift heuristic, and an exhaustive finite verification through 10^9.

The present paper asks a structural question rather than performing a new large search. If both subtraction constants are multiplied by a common power 4^r, and the state space is simultaneously scaled by 4^r, does a genuinely new dynamical system appear? We prove that the answer is no. The entire family is exactly conjugate to the base map through the linear scaling $y \mapsto 4^r y$.

This distinction is important. Numerical similarity at several small values of r would only suggest a pattern. Exact conjugacy proves the pattern for every integer $r \geq 0$ at once. It also specifies precisely what transfers: complete trajectories, cycle lengths, branch choices, 3-adic valuations, stopping/transient times, basin membership, and normalized orbit geometry. Absolute state values and peaks scale by 4^r.

The finite computation of [2] remains base-system evidence. Its numerical counts are cited here only as inherited data and are not presented as a new billion-scale run of every scaled system. The new contribution is algebraic: a theorem identifying all scaled systems with the same base dynamics.

2 The base ternary map

Let

$$\mathcal{S} = \{y \in \mathbb{Z}_{>0} : 3 \nmid y\}. \tag{1}$$

For $y \in \mathcal{S}$, define

$$\rho(y) = y \bmod 3 \in \{1, 2\}.$$

The base affine pre-reduction is

$$M_0(y) = 4y - \rho(y) = \begin{cases} 4y - 1, & y \equiv 1 \pmod 3, \\ 4y - 2, & y \equiv 2 \pmod 3. \end{cases} \tag{2}$$

Because $4 \equiv 1 \pmod 3$, the quantity $M_0(y)$ is always divisible by 3.

For a nonzero integer n, let

$$\nu_3(n) = \max\{j \geq 0 : 3^j \mid n\}.$$

The accelerated base map is

$$T_0(y) = \frac{M_0(y)}{3^{\nu_3(M_0(y))}}. \tag{3}$$

The output again belongs to $\mathcal{S}$.

Writing $y = 3q + 1$ or $3q + 2$ gives

$$M_0(3q+1) = 3(4q+1), \qquad M_0(3q+2) = 3(4q+2).$$

The positive attractors observed and exhaustively verified through the finite base range are the fixed points 1, 2, and the four-cycle

$$22 \longrightarrow 29 \longrightarrow 38 \longrightarrow 50 \longrightarrow 22. \tag{4}$$

3 The 4^r-scaled family

Fix an integer $r \geq 0$.

3.1 Scaled domain

Define

$$\mathcal{S}_r = 4^r\mathcal{S} = \{4^r y : y \in \mathcal{S}\}. \tag{5}$$

Every $x \in \mathcal{S}_r$ has a unique representation $x = 4^r y$ with $y \in \mathcal{S}$.

Since $4 \equiv 1 \pmod 3$,

$$4^r \equiv 1 \pmod 3, \tag{6}$$

so

$$x = 4^r y \equiv y \pmod 3. \tag{7}$$

Thus scaling preserves the nonzero residue class modulo 3.

3.2 Scaled affine and compressed maps

Define

$$M_r(x) = 4x - 4^r\rho(x), \qquad x \in \mathcal{S}_r, \tag{8}$$

where $\rho(x) = x \bmod 3 \in \{1, 2\}$. Equivalently,

$$M_r(x) = \begin{cases} 4x - 4^r, & x \equiv 1 \pmod 3, \\ 4x - 2 \cdot 4^r, & x \equiv 2 \pmod 3. \end{cases} \tag{9}$$

The corresponding accelerated map is

$$T_r(x) = \frac{M_r(x)}{3^{\nu_3(M_r(x))}}. \tag{10}$$

Well-definedness. If $x \equiv 1 \pmod 3$, then $4x - 4^r \equiv 1 - 1 \equiv 0 \pmod 3$. If $x \equiv 2 \pmod 3$, then $4x - 2 \cdot 4^r \equiv 2 - 2 \equiv 0 \pmod 3$. Hence $\nu_3(M_r(x)) \geq 1$ for every $x \in \mathcal{S}_r$.

4 The key 3-adic scaling identity

Lemma 4.1 (Affine scaling). For every $r \geq 0$ and $y \in \mathcal{S}$,

$$M_r(4^r y) = 4^r M_0(y). \tag{11}$$

Proof. By (7), $\rho(4^r y) = \rho(y)$. Therefore

$$M_r(4^r y) = 4(4^r y) - 4^r\rho(y) = 4^r(4y - \rho(y)) = 4^r M_0(y).$$

□

Lemma 4.2 (Valuation invariance). For every nonzero integer n and every $r \geq 0$,

$$\nu_3(4^r n) = \nu_3(n). \tag{12}$$

Proof. Since $\gcd(4^r, 3) = 1$, the factor 4^r contributes no factor of 3. □

Combining the two lemmas gives

$$\nu_3(M_r(4^r y)) = \nu_3(M_0(y)). \tag{13}$$

This equality is stronger than mere orbit scaling: corresponding trajectories remove exactly the same power of 3 at every matched step.

5 Exact algebraic conjugacy

Define the scaling bijection

$$\Phi_r : \mathcal{S} \to \mathcal{S}_r, \qquad \Phi_r(y) = 4^r y. \tag{14}$$

Its inverse is $\Phi_r^{-1}(x) = x/4^r$.

Theorem 5.1 (One-step conjugacy). For every $r \geq 0$ and $y \in \mathcal{S}$,

$$\boxed{T_r(4^r y) = 4^r T_0(y).} \tag{15}$$

Equivalently,

$$T_r \circ \Phi_r = \Phi_r \circ T_0, \qquad T_r = \Phi_r \circ T_0 \circ \Phi_r^{-1}. \tag{16}$$

Proof. From (11) and (13),

$$T_r(4^r y) = \frac{4^r M_0(y)}{3^{\nu_3(M_0(y))}} = 4^r T_0(y).$$

□

Theorem 5.2 (All iterates). For every $j \geq 0$, $r \geq 0$, and $y \in \mathcal{S}$,

$$\boxed{T_r^j(4^r y) = 4^r T_0^j(y).} \tag{17}$$

Proof. The case $j = 0$ is immediate. If the identity holds for j, then

$$T_r^{j+1}(4^r y) = T_r(4^r T_0^j(y)) = 4^r T_0(T_0^j(y)) = 4^r T_0^{j+1}(y),$$

using Theorem 5.1. □

6 Exact preservation of cycles

Theorem 6.1 (Cycle bijection). A tuple

$$C = (c_0, c_1, \ldots, c_{L-1})$$

is a cycle of T_0 if and only if

$$4^r C = (4^r c_0, 4^r c_1, \ldots, 4^r c_{L-1})$$

is a cycle of T_r. Minimal cycle length is preserved.

Proof. If $T_0^L(c_0) = c_0$, then Theorem 5.2 gives

$$T_r^L(4^r c_0) = 4^r T_0^L(c_0) = 4^r c_0.$$

Conversely, if $x = 4^r y \in \mathcal{S}_r$ satisfies $T_r^L(x) = x$, then

$$4^r T_0^L(y) = T_r^L(4^r y) = 4^r y,$$

so $T_0^L(y) = y$. Since Φ_r is bijective, a shorter period on one side would imply the same shorter period on the other. □

Corollary 6.2. No cycle exists in $\mathcal{S}_r$ that is not the 4^r-multiple of a base cycle. Thus scaling introduces no new periodic structure inside the scaled domain.

For the known positive attractors, the fixed points become

$$4^r \longrightarrow 4^r, \qquad 2 \cdot 4^r \longrightarrow 2 \cdot 4^r, \tag{18}$$

and the four-cycle becomes

$$22 \cdot 4^r \to 29 \cdot 4^r \to 38 \cdot 4^r \to 50 \cdot 4^r \to 22 \cdot 4^r. \tag{19}$$

For example, at $r = 1$ the attractors are 4, 8, and

$$88 \to 116 \to 152 \to 200 \to 88,$$

while at $r = 2$ they are 16, 32, and

$$352 \to 464 \to 608 \to 800 \to 352.$$

These examples are exact consequences of the theorem, not independent numerical discoveries.

7 The cycle equation under scaling

The base one-step cycle relation from [2] is

$$3^{k_i} y_{i+1} = 4y_i - \rho_i, \qquad \rho_i \in \{1, 2\}, \quad k_i \geq 1. \tag{20}$$

For the scaled system it becomes

$$3^{k_i} x_{i+1} = 4x_i - 4^r \rho_i. \tag{21}$$

Putting $x_i = 4^r y_i$ and dividing by 4^r recovers (20) exactly. Hence the scaled cycle equation contains no new Diophantine condition.

Let $K = \sum_{i=0}^{L-1} k_i$. Iterating (21) around a length-L cycle gives

$$x_0 = 4^r \, \frac{\sum_{j=0}^{L-1} \rho_j 4^{L-1-j} 3^{\sum_{m=0}^{j-1} k_m}}{4^L - 3^K}. \tag{22}$$

The fraction is exactly the base starting value y_0. Therefore

$$\boxed{x_0 = 4^r y_0.} \tag{23}$$

This gives a second, direct algebraic proof of cycle preservation from the closed Diophantine equation.

For the base four-cycle, $L = 4$, $K = 5$, residues $(1, 2, 2, 2)$, and valuations $(1, 1, 1, 2)$. The denominator is

$$4^4 - 3^5 = 256 - 243 = 13,$$

and the base numerator is 286, yielding $y_0 = 22$. The scaled numerator is $4^r \cdot 286$, so $x_0 = 22 \cdot 4^r$.

8 Orbit statistics preserved by conjugacy

Let $y_j = T_0^j(y)$ and $x_j = T_r^j(4^r y)$. By Theorem 5.2,

$$x_j = 4^r y_j \quad \text{for every } j \geq 0. \tag{24}$$

Several consequences are immediate.

8.1 Transient length and valuation totals

If $\tau_0(y)$ is the first iteration at which the base orbit enters its eventual cycle, then

$$\tau_r(4^r y) = \tau_0(y). \tag{25}$$

Because the matched 3-adic valuations are equal step by step, the total number of individual divisions by 3 before cycle entry is also identical.

8.2 Peaks and normalized ratios

For any matched finite trajectory segment,

$$\max_j x_j = 4^r \max_j y_j. \tag{26}$$

Thus absolute reduced peaks scale by 4^r. The same is true for the raw affine peak because $M_r(4^r y) = 4^r M_0(y)$. On the other hand,

$$\frac{x_j}{x_0} = \frac{y_j}{y_0}, \tag{27}$$

so every normalized peak ratio is invariant.

8.3 Basin membership

For any base cycle C, let $\mathcal{B}_0(C)$ be its basin and $\mathcal{B}_r(4^r C)$ the corresponding scaled basin. Then

$$4^r y \in \mathcal{B}_r(4^r C) \iff y \in \mathcal{B}_0(C). \tag{28}$$

Therefore basin counts are exactly preserved on corresponding finite sets. They should not be compared at the same absolute cutoff unless the different domain sizes are accounted for.

9 The 3-adic drift viewpoint

The base paper derives the heuristic distribution

$$\Pr(\nu_3 = k) = \frac{2}{3^k}, \qquad k \geq 1, \tag{29}$$

with

$$\mathbb{E}[\nu_3] = \frac{3}{2}.$$

The associated average logarithmic drift is

$$\Delta = \log 4 - \frac{3}{2}\log 3 \approx -0.261624, \tag{30}$$

and the corresponding geometric factor is

$$\lambda = \frac{4}{3\sqrt{3}} \approx 0.7698. \tag{31}$$

For matched states,

$$\frac{T_r(4^r y)}{4^r y} = \frac{T_0(y)}{y}, \tag{32}$$

so the scaled family has exactly the same normalized one-step ratios and therefore inherits the same heuristic drift model. As in the base paper, this probabilistic heuristic is not a proof of global convergence.

10 Finite verification inherited from the base paper

The base computation in [2] exhaustively tested every admissible start

$$1 \leq y \leq 10^9, \qquad 3 \nmid y.$$

There were exactly $666,666,667$ such starts, with zero reported overflow. The finite basin counts were:

Table 1: Base-system finite verification through 10^9, inherited from [2].

Attractor	Count	Percentage
Fixed point 1	66,019,616	9.902942395%
Fixed point 2	432,100,455	64.815068218%
$22 \to 29 \to 38 \to 50 \to 22$	168,546,596	25.281989387%
Total	666,666,667	100%

The longest transient had 405 reduced iterations, beginning at $833,560,375$, and the maximum accumulated number of divisions by 3 was 529, attained by the same start. The largest reduced state observed was

$$164,210,561,845,359,809,$$

from start $664,934,564$.

These are not new measurements of the scaled family. The new theorem implies that, for every fixed r, the corresponding $666,666,667$ scaled starts

$$\{4^r y : 1 \leq y \leq 10^9,\ 3 \nmid y\} \tag{33}$$

are verified algebraically without re-running the dynamical search. They lie in the absolute interval up to $4^r 10^9$, restricted to $\mathcal{S}_r$. Their basin counts and transient lengths are identical, while every state peak is multiplied by 4^r.

For example, the largest reduced peak among this corresponding scaled set is

$$4^r \cdot 164,210,561,845,359,809, \tag{34}$$

and it occurs for the scaled start

$$4^r \cdot 664,934,564. \tag{35}$$

Likewise, the 405-iteration record corresponds to start $4^r \cdot 833,560,375$.

11 Finite theorem and global conjecture

The conjugacy allows a precise separation between an exact theorem and the unresolved global statement.

Finite transfer theorem. For every $r \geq 0$, every scaled start $x = 4^r y$ with $1 \leq y \leq 10^9$ and $3 \nmid y$ enters one of

$$4^r, \qquad 2 \cdot 4^r, \qquad 22 \cdot 4^r \to 29 \cdot 4^r \to 38 \cdot 4^r \to 50 \cdot 4^r \to 22 \cdot 4^r.$$

This follows rigorously from the exhaustive finite base computation together with Theorem 5.2.

Conjecture (scaled three-attractor convergence). For every $r \geq 0$ and every $x \in \mathcal{S}_r$, the orbit of x under T_r eventually enters one of the two scaled fixed points or the scaled four-cycle above.

Theorem 11.1 (Equivalence of global conjectures). The scaled three-attractor conjecture holds for any one $r \geq 0$ if and only if the base three-attractor conjecture holds on $\mathcal{S}$. Consequently it holds for all $r \geq 0$ if and only if it holds for $r = 0$.

Proof. Conjugacy transports every orbit in both directions under the bijection Φ_r. □

Thus the parameter r creates no additional global convergence problem. It also does not solve the base problem: an unbounded base trajectory, if one exists, would scale to an unbounded trajectory at every level, and an undiscovered base cycle would scale to a corresponding cycle at every level.

12 What is proved and what is not

The exact algebraic results of this paper prove that, for every $r \geq 0$:

(i) $\mathcal{S}_r = 4^r\mathcal{S}$ is in bijection with the base domain;

(ii) multiplication by 4^r preserves the relevant residue modulo 3;

(iii) the scaled affine numerator satisfies $M_r(4^r y) = 4^r M_0(y)$;

(iv) the complete 3-adic valuation is unchanged at corresponding steps;

(v) T_r and T_0 are exactly conjugate;

(vi) all iterates correspond exactly;

(vii) periodic orbits and minimal cycle lengths correspond bijectively;

(viii) valuation sequences and transient lengths are preserved;

(ix) absolute peaks scale by 4^r, while normalized ratios are preserved;

(x) basin membership transfers exactly;

(xi) the entire finite base verification through 10^9 transfers to the corresponding scaled set through absolute scale $4^r 10^9$.

What is not proved is global convergence of every positive base orbit. The finite computation through 10^9 cannot exclude a new base cycle or an unbounded base trajectory lying beyond the tested region. Since the scaled systems are conjugate to the base system, they inherit exactly this unresolved status.

13 Reproducibility and computational verification of the identity

The supplementary archive accompanying this paper contains the LaTeX source, compiled PDF, a Python verifier, a machine-readable JSON statement of the scaling rules, a README, and the base 10^9 global summary copied as a clearly labelled inherited reference. The verifier checks, for user-selected finite ranges of r and y, that

$$M_r(4^r y) = 4^r M_0(y), \quad \nu_3(M_r(4^r y)) = \nu_3(M_0(y)), \quad T_r(4^r y) = 4^r T_0(y),$$

and can also compare several iterates.

This small verifier is a consistency check, not the proof. The proof is the algebraic factorization in Lemma 4.1 together with $\gcd(4^r, 3) = 1$. The billion-scale numerical evidence belongs to the base record [2]; the present work deliberately avoids relabelling inherited numerical data as a new computation.

14 Conclusion

For the ternary $(4k-1(2))/3$ Collatz-type map, multiplying both branch subtraction constants and the state space by 4^r produces an exact scaled copy of the base dynamics. The generalized branches are

$$4x - 4^r \quad \text{and} \quad 4x - 2 \cdot 4^r,$$

and on $\mathcal{S}_r = 4^r\mathcal{S}$ the compressed map satisfies

$$\boxed{T_r(4^r y) = 4^r T_0(y).}$$

The identity holds because powers of 4 are simultaneously congruent to 1 modulo 3 and coprime to 3. The first property preserves branch selection; the second preserves the complete 3-adic valuation.

The consequences are exact rather than heuristic. Every scaled trajectory is the 4^r-multiple of one base trajectory. Every scaled cycle is the 4^r-multiple of one base cycle and has the same length. In particular, the two fixed points and the four-cycle of the base finite verification become their scaled counterparts, and no additional periodic orbit can exist inside a scaled domain unless a corresponding base periodic orbit exists.

The exhaustive base verification through 10^9 therefore transfers without a new large computation to every corresponding 4^r-scaled finite domain. At the same time, the global convergence question is unchanged: proving the three-attractor conjecture for the base system would prove it simultaneously for the whole scaled family, while any counterexample in the base system would propagate throughout the family. The 4^r parameter is thus an exact scaling symmetry, not a source of independent dynamics.

References

[1] J. C. Lagarias, "The $3x+1$ problem and its generalizations," *The American Mathematical Monthly*, 92(1), 3–23, 1985. DOI: 10.2307/2322189.

[2] F. Banazadeh, "A Ternary $(4k-1(2))/3$ Collatz-Type Map: Complete 3-Adic Reduction, Algebraic Cycle Analysis, and Exhaustive Computational Verification up to 10^9," Zenodo, 2026. DOI: 10.5281/zenodo.22662980.

[3] F. Banazadeh, "Sequence Computations," GitHub repository, 2026. `https://github.com/FarhadBanazadeh/sequence-computations`.